

Assignment Number: 3.3

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BATCH: 24

Task 1: AI-Generated Logic for Reading Consumer Details

Scenario

An electricity billing system must collect accurate consumer data.

Task Description

Use an AI tool (GitHub Copilot / Gemini) to generate a Python program that:

- Reads:
 - o Previous Units (PU)
 - o Current Units (CU)
 - o Type of Customer
- Calculates units consumed
- Implements logic directly in the main program (no functions)
- Expected Output
- Correct input reading
- Units consumed calculation

```
53 # application for TGNPDCL-electricity bill generation take 3 user inputs,
54 pervious_inputs = float(input("Enter Previous Month's Meter Reading: "))
55 current_inputs = float(input("Enter Current Month's Meter Reading: "))
56 units_consumed = current_inputs - pervious_inputs
57 customer_type = input("Enter Customer Type (Domestic/Commercial/Industrial) ")
58 if current_inputs < pervious_inputs:
59     print("Error: Current reading cannot be less than previous reading.")
60     exit()
61 units_consumed = current_inputs - pervious_inputs
62 print("Units Consumed:", units_consumed)
63
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Enter Previous Month's Meter Reading: 20
Enter Current Month's Meter Reading: 35
Enter Previous Month's Meter Reading: 20
Enter Current Month's Meter Reading: 35
Enter Customer Type (Domestic/Commercial/Industrial): commercial
Enter Customer Type (Domestic/Commercial/Industrial): commercial
Units Consumed: 15.0
Units Consumed: 15.0

- Screenshot showing AI-generated code
- Sample input and output

Task 2: Energy Charges Calculation Based on Units Consumed

Scenario

Energy charges depend on the number of units consumed and customer type.

Task Description

Review the AI-generated code from Task 1 and extend it to:

- Calculate Energy Charges (EC)
- Use conditional statements based on:

o Domestic

o Commercial

o Industrial consumers

• Improve readability using AI prompts such as:

o “Simplify energy charge calculation logic”

o “Optimize conditional statements”

Expected Output

• Correct EC calculation

• Clear conditional logic

• Original and improved versions (optional)

• Sample execution results

```
65 #write a function for Energy charges calculation based on customer type, and optimal code
66 def calculate_energy_charges(units_consumed, customer_type):
67     if customer_type.lower() == "domestic":
68         if units_consumed <= 100:
69             charges = units_consumed * 1.5
70         elif units_consumed <= 300:
71             charges = (100 * 1.5) + ((units_consumed - 100) * 2.5)
72         else:
73             charges = (100 * 1.5) + (200 * 2.5) + ((units_consumed - 300) * 4.0)
74     elif customer_type.lower() == "commercial":
75         if units_consumed <= 100:
76             charges = units_consumed * 2.0
77         elif units_consumed <= 300:
78             charges = (100 * 2.0) + ((units_consumed - 100) * 3.5)
79         else:
80             charges = (100 * 2.0) + (200 * 3.5) + ((units_consumed - 300) * 5.0)
81     elif customer_type.lower() == "industrial":
82         if units_consumed <= 100:
83             charges = units_consumed * 3.0
84         elif units_consumed <= 300:
85             charges = (100 * 3.0) + ((units_consumed - 100) * 4.5)
86         else:
87             charges = (100 * 3.0) + (200 * 4.5) + ((units_consumed - 300) * 6.0)
88     else:
89         raise ValueError("Invalid customer type")
90     return charges
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Enter Previous Month's Meter Reading: 34.6
Enter Current Month's Meter Reading: 45
Enter Customer Type (Domestic/Commercial/Industrial): commercial
Enter Current Month's Meter Reading: 45
Enter Current Month's Meter Reading: 45
Enter Customer Type (Domestic/Commercial/Industrial): commercial
Units Consumed: 10.399999999999999
Energy Charges for commercial customer: 20.799999999999997
```

Task 3: Modular Design Using AI Assistance (Using Functions)

Scenario

Billing logic must be reusable for multiple consumers.

Task Description

Use AI assistance to generate a Python program that:

- Uses user-defined functions to:
 - o Calculate Energy Charges
 - o Calculate Fixed Charges
- Returns calculated values
- Includes meaningful comments

Expected Output

- Function-based Python program
- Correct EC and FC values
- Screenshots of AI-assisted function generation
- Test cases with output

```
94 #write a well-commented python optimal code to calculate fixed charges based on consumed units and customer typ
95 def calculate_fixed_charges(units_consumed, customer_type):
96     # Define fixed charges based on customer type
97     fixed_charges = 0
98     if customer_type.lower() == "domestic":
99         if units_consumed <= 100:
100             fixed_charges = 50
101         elif units_consumed <= 300:
102             fixed_charges = 100
103         else:
104             fixed_charges = 150
105     elif customer_type.lower() == "commercial":
106         if units_consumed <= 100:
107             fixed_charges = 100
108         elif units_consumed <= 300:
109             fixed_charges = 200
110         else:
111             fixed_charges = 300
112     elif customer_type.lower() == "industrial":
113         if units_consumed <= 100:
114             fixed_charges = 150
115         elif units_consumed <= 300:
116             fixed_charges = 300
117         else:
118             fixed_charges = 450
119     return fixed_charges
120 fixed_charges = calculate_fixed_charges(units_consumed, customer_type)
121 print("Fixed Charges for", customer_type, "customer:", fixed_charges)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Enter Previous Month's Meter Reading: 45
Enter Current Month's Meter Reading: 356
Enter Customer Type (Domestic/Commercial/Industrial): domestic
Units Consumed: 311.0
Energy Charges for domestic customer: 694.0
Fixed Charges for domestic customer: 150
```

Task 4: Calculation of Additional Charges

Scenario

Electricity bills include multiple additional charges.

Task Description

Extend the program to calculate:

- FC – Fixed Charges
- CC – Customer Charges
- ED – Electricity Duty (percentage of EC)

Use AI prompts like:

- “Add electricity duty calculation”
- “Improve billing accuracy”

Expected Output

- Individual charge values printed
- Correct duty calculation
- Well-structured output
- Verified intermediate results

```
123
124 #extend the above code to calculate customer charges and add electricity duty calculation to improve billing accuracy without creating functions
125 customer_charges = 20
126 electricity_duty = 0.05
127 total_charges = charges + fixed_charges + customer_charges
128 electricity_duty_amount = total_charges * electricity_duty
129 print("Total Charges:", total_charges)
130 print("Electricity Duty Amount:", electricity_duty_amount)
131 print("Final Bill Amount:", total_charges + electricity_duty_amount)
132
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\yadav\OneDrive\Desktop\AI-LAB> & C:/Users/yadav/AppData/Local/Microsoft/windowsApps/python3.13.exe c:/Users/yadav/OneDrive/Desktop/AI-LAB/assign.py
Enter Previous Month's Meter Reading: 20
Enter Current Month's Meter Reading: 30
Enter Customer Type (Domestic/Commercial/Industrial): commercial
Units Consumed: 10.0
Energy Charges for commercial customer: 20.0
Fixed Charges for commercial customer: 100
Total Charges: 140.0
Electricity Duty Amount: 7.0
Final Bill Amount: 147.0
PS C:\Users\yadav\OneDrive\Desktop\AI-LAB>
```

Task 5: Final Bill Generation and Output Analysis

Scenario

The final electricity bill must present all values clearly.

Task Description

Develop the final Python application to:

- Calculate total bill:
- $\text{Total Bill} = \text{EC} + \text{FC} + \text{CC} + \text{ED}$
- Display:
 - o Energy Charges (EC)
 - o Fixed Charges (FC)
 - o Customer Charges (CC)
 - o Electricity Duty (ED)
 - o Total Bill Amount
- Analyze the program based on:
 - o Accuracy
 - o Readability
 - o Real-world applicability

Expected Output

- Complete electricity bill output
- Neatly formatted display
- Sample input/output
- Short analysis paragraph

```
134 #calculate the final bill amount by adding energy charges, fixed charges, customer charges and electricity duty
135 final_bill_amount = total_charges + electricity_duty_amount+ customer_charges+ fixed_charges
136 #display energy charges, fixed charges, customer charges, electricity duty and final bill amount
137 print("Energy Charges:", charges)
138 print("Fixed Charges:", fixed_charges)
139 print("Customer Charges:", customer_charges)
140 print("Electricity Duty Amount:", electricity_duty_amount)
141 print("Final Bill Amount:", final_bill_amount)
142
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\yadav\OneDrive\Desktop\AI-LAB> & C:/Users/yadav/AppData/Local/Microsoft/WindowsApps/python3.13.exe c:/Users/yadav/OneDr
Enter Previous Month's Meter Reading: 30
Enter Current Month's Meter Reading: 40
Enter Customer Type (Domestic/Commercial/Industrial): commercial
Units Consumed: 10.0
Energy Charges for commercial customer: 20.0
Fixed Charges for commercial customer: 100
Total Charges: 140.0
Electricity Duty Amount: 7.0
Final Bill Amount: 147.0
Energy Charges: 20.0
Fixed Charges: 100
Customer Charges: 20
Electricity Duty Amount: 7.0
Final Bill Amount: 267.0
PS C:\Users\yadav\OneDrive\Desktop\AI-LAB> |
```